MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY:10S4S483

(FIRING INDICATOR)

10 SCOPE

1.1 This specification covers the detail requirements for one type of circuit card asseubly. The requirements of this specification together with the general requirements of specification MIL-C-S0785 comprise the complete requirements for the Firing Indicator Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-5078S	Circuit Card *semblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

1054S42S	Sight Current Generator
10S4S483	Circuit *Card Assembly,* Firing Indicator

PACKAGING DATA SHEET

1054S483	Packaging of Circuit Card Assetily, Firing Indicator

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item definition. This assembly is a single ended printed circuit board consisting of diodes, capacitors and resistors so arranged electrically as to provide the following function.

a. Free running multivibrator
b. Switching network
c. Too (2] AND gates
d. No (2) lamp drivers
e. Precision voltage divider

The assembly controls the READY TO FIRE and the GOOD WHEN LIT control panel indicator lights when it is installed in the Sight Current Generator, Drawing Number 10S45425.

3.1.1 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL) 10545483.

3.1.2 Government furnished property list. Not applicable,

3.2 Characteristics

3.2.1 Performance. The assembly shall be capable of performing the following functions when assembled in accordance with Drawing Number 10S4S483.

a. Free running multivibrator shall provide a 2.5 Hz 6 volt p-P output.

b. Switching network shall control the application of the multivibrator output to the Q4 and Q7 lamp driver circuit.

c. Six-input AND gate shall control the Q4 and Q7 lamp driver circuit.

d. Three-input AND gate shall control the Q6 and Q8 lamp driver circuit.

e. No lamp driver circuits shall control the READY TO FIRE and GOOD WHEN LIT indicator lights in the system control panel.

f. Precision voltage divider shall provide two reference voltages for use in the GOOD WHEN LIT circuitry.

3.2.1.1 Input and Output levels

3.2.1.1.1 Input power. The input power supplied to the assembly shall be as specified in Table 1.

Table I. Input Power

Pin	Voltage
4 and 25	+15 vdc ± 2.0 percent
14	-15 vdc ± 2.0 percent
24	+24 vdc ± 2.0 percent
1 and 28	Ground
18	+24 vdc ± 2.0 percent
27	+24 vdc ± 2.0 percent

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3,2,1.1.2 The requirements specified in 3.2.1.1.2.1 through 3.2.1,1.2.4 s all e met when supplied with input voltages specified in Table I.

3,2.1.1.2.1 Pin 18 output, The output between pin 18 and pin 28 shall be as follows .

a. With pins 2, 3, 5, 6, 8, 9 and 13 open the output shall be less than 500 millivolts.

b. with one or more *of* pins 3, 5, 6, 8, and 9 grounded or with 15 Vdc applied between pin 2 and pin one, and with pin 13 open the output shall be 24 Vdc & 5.0 percent.

c. With pin 13 grounded and pins 3, 5, 6, 8 and 9 either open or grounded, and pin 2 either open or connected to +15 Vdc, the output shall be a rectangular wave alternating between 24 Vdc ± 5.0 percent and less than 500 millivolts, with a time period of 320 to 480 milliseconds and a duty cycle of 40 *to* 60 percent.

3.2.1.1.2.2 Pin 19 output. The output shall be +3.90 Vdc ± 2.0 percent between pin 19 and pin 28.

3,2.1,1.2.3 Pin 20 output. The output shall be +3.70 Vdc ± 2.0 percent between pin 20 and pin 28.

3.2.1.1.2.4 Pin 27 output. The output shall be less than 500 millivolts between pin 27 and pin 28 when pins 21, 22 and 23 are open, and 24 Vdc ± 5.0 percent with one or more of pins 21, 22, and 23 grounded.

3.2.1.2 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic. The physical characteristics shall be as specified in drawing 10S45483 and the engineering parts list.

3.3 Design and construction

3.3.1 Production drawings. This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 10S45483 and associated lists.

3.3.2 Standards of manufacture. The standards of manufacture shall be those specified in specification MIL-C-S0785.

3.3.3 Workmanship. The workmanship shall be as specified in specification MIL-C-S078S.

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4. QUALITY ASSURANCE PROVISION

4.1 General

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-S0785.

4.1.2 Special tests and examination. lhe assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-S0785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment

4.2.1.1 Power supply (PSl) The preparer supply shall be is ± 0.1 Vdc, 100 milliamperes or greater, ripple less than 10 millivolts (Mv) rms or 100 mv peak-to-peak.

4.2,1.2 Power supply (PSZ). The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes or greater, ripple less than 10 mv rms or 100 mw peak-to-peak.

4.2.1.3 Power supply (Ps3). The power supply shall be 24 2 0.5 Vdc, 500 milliamperes or greater,

4.2.1.4 Lamps. Two lamps are required with a rating of 0.18 amperes nominal at +24 Vdc.

4.2.1.S Switches. Ten single pole single throw switches are required.

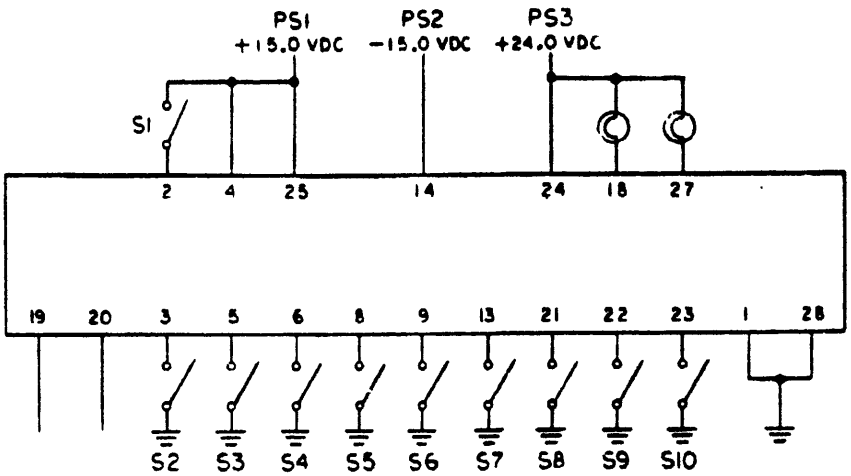
4.2.1.6 Digital voltmeter. The digital voltmeter (DVM) shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 Vdc. Input impedance greater than 10 megohms on all voltage scales.

4.2.1.7 Oscilloscope. The oscilloscope shall have a voltage and time accuracy and resolution o 5.0 percent or better, with a rise the of 100 nanoseconds or better, and with automatic triggering. Voltage sensitivity shall include the range of 0.05 volt/cm to 5.0 volt/cm.

4.2.2 Functional tests

4.2.2.1 Test connections. The power supplies, lamps and switches shall be connetted to the assembly under test in accordance with Figure 1.

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Note: All switches shown in the OFF position.

Figure 1. Test Connections

4.2.2.2 Test procedure. Compliance with the requirements of 3.2.1 .1.2 shall be verified as follows:

a. With all switches S1 through S7 OFF, the voltage at pin 18 measured with the DVM shall be less than 500 millivolts dc.

b. With one or more of switches S1 through S6 ON and S7 OFF, the voltage at pin 18 shall be 24 Vdc ± 5.0 percent.

c* With switch S7 ON and switches S1 through S6 either ON or OFF, the voltage at pin 18, measured with the oscilloscope, shall be the rectangular wave described in 3.2.1.1.2.lc.

d. With switches S8, S9 and S10 OFF, the voltage at pin 27 measured with the DVM shall be less than 500 millivolts dc.

e. The ratio of the Vdc at pin 19 to the Vdc at pin 4 shall be between 0.2573 and 0.2626.

f. The ratio of the Vdc at pin 20 to the Vdc at pin 4 shall be between 0.2440 and 0.2490.

g. With one or more of switches S8, S9, and SIG uN, the voltage at pin 27 shall be 24 Vdc ± 5.0 percent.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC form 1029, 10545485.

6, NOTES:

6.1 The notes specified in the general specification MIL-C-S078S are applicable to this specification.

Custodian

Army - MU

PrepARing Activity:

Army - MU

Project No. 1285-A309

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Print or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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